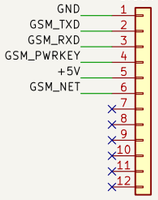
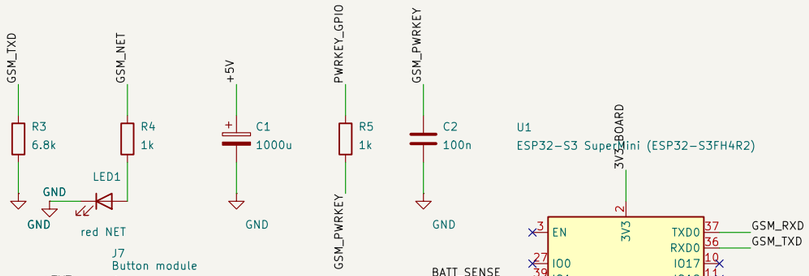
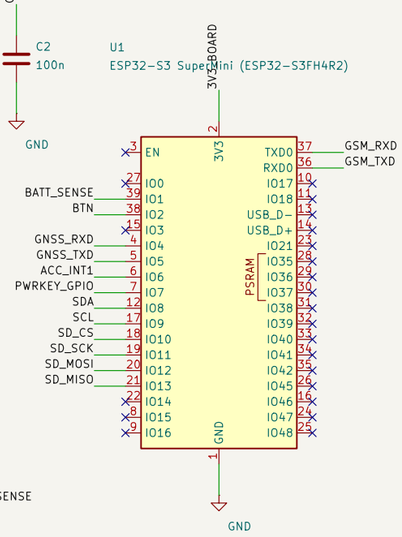


1 VCC 5V 2 GND 3 TX -> IO5 4 RX <- IO4
5 SCL / 6 SDA: compass (Foxeer only, 0x0D). BE-220 has pins 1-4.

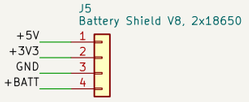




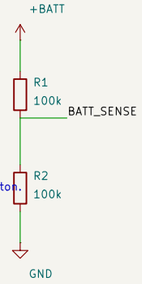
1 VCC 3V3 2 GND 3 SIG -> IO2
Own pull-down: SIG low at rest,
3V3 when pressed (buttonPressedHigh).



Silkscreen: GND TX (5 V level!) -> IO44 RX <- IO43 PK = BOOT/PWRKEY VSS = VCC 5V in NET
6 NET -> R4 1k -> LED1 (NETLIGHT: 0.8 s searching, 3 s registered, 0.3 s GPRS) 7 DTR 8 RING 9-12 audio: unused.
TX high side is pulled to VCC through 4.7k on the board: R3 divides it to ~3.0 V for IO44.
PK <- R5 1k <- IO7 (open drain). C2 100n PK to GND, at the ESP end: a bare wire carried every attach burst in.
VBAT is 3.64 V (MIC29302). C1 1000 uF across VSS/GND right at the board's pins (+ to VSS): the onboard
100+10+1 uF are not enough on battery (reset every 30 s during registration without it). Short thick wires.

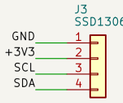


1 5V out (<= 2.2 A cont.) 2 3V out (1 A; measure: 3.0 or 3.3 V) 3 GND 4 B+ (cell)
Feeds: +5V -> SuperMini 5V pin (USB unplugged), GNSS, SIM800C board, SD. +3V3 -> OLED, BMA400, button.

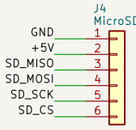


Cell 4.2 V max -> 2.1 V at IO1 (ADC1_CH0, 12 dB).

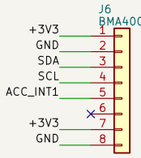
SuperMini: 5V pin = its USB VBUS (no diode): +5V from the shield only with USB unplugged.
Onboard LDO makes 3V3_BOARD. USB (console), EN, BOOT and the RGB LED (IO48) stay on the board.
DevKitC-1 build: GSM on IO17/IO18 instead of IO43/IO44.



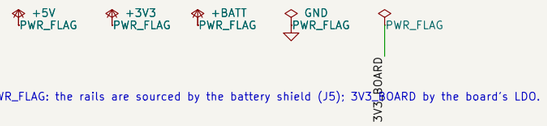
1 GND 2 VCC 3V3 3 SCL <- IO9 4 SDA <> IO8
Module pull-ups on SDA/SCL; bus at 100 kHz.



1 GND 2 VCC 5V (adapter has its own 3.3 V regulator and level shifter)
3 MISO -> IO13 4 MOSI <- IO12 5 SCK <- IO11 6 CS <- IO10



1 VCC 3V3 2 GND 3 SDA 4 SCL 5 INT1 -> IO6 (RTC GPIO, deep-sleep wake) 6 INT2 n/c
7 CS -> 3V3: I2C mode at power-up (floating, the chip started in SPI mode on 1 boot in 3)
8 SDO -> GND: address 0x14 (0x15 with SDO high; the firmware probes both).



PWR_FLAG: the rails are sourced by the battery shield (J5); 3V3_BOARD by the board's LDO.

Breakout modules drawn as pin headers; see hardware/README.md
ESP32-S3 SuperMini build; pins per firmware/include/config.hpp
Maksym Kutyshenko

Sheet: /
File: gps_tracker.kicad_sch

Title: GPS Tracker

Size: A3	Date: 2026-10-15	**Rev: 2**
KiCad E.D.A. 10.0.4		Id: 1/1